100 Days of Linux - Day 048

Title: Managing Processes with kill

Today's Goal
Learn how to stop running processes safely using the **kill** command and understand common Linux signals.

Why This Matters
Developers, system administrators, and data engineers often stop stuck scripts, failed jobs, or unnecessary processes to free system resources.

Command of the Day: kill

Common Syntax:
kill PID
kill -15 PID (SIGTERM - graceful stop)
kill -9 PID (SIGKILL - force stop)
kill -l (list available signals)

Common Signals:
15 (SIGTERM) = Request the process to stop gracefully
9 (SIGKILL) = Force the process to stop immediately

Tip: Use **ps**, **top**, or **htop** to find the PID before using **kill**.

Beginner Lesson
Every running process has a unique Process ID (PID). The **kill** command sends a signal to that process. Start with SIGTERM whenever possible because it allows the program to close cleanly. Use SIGKILL only if the process refuses to stop.

Practice Questions

1. Open a new terminal and run the command 'sleep 300' to create a long-running process.
2. Use ps or htop to find the PID of the sleep process.
3. Stop the sleep process using kill with the default signal (SIGTERM). Verify that it has stopped.
4. Start another sleep 300 process. Stop it using kill -9 and verify that it is no longer running.
5. Display your current working directory after completing today's tasks.

Hints

Always identify the correct PID before sending a signal. Verify the process has stopped by running ps again.

Mini Challenge

Run two 'sleep 300' processes. Stop one with SIGTERM and the other with SIGKILL. Compare the commands you used and verify both processes are gone.

Common Mistakes
Using kill -9 immediately. Try the default kill command (SIGTERM) first because it allows programs to exit gracefully.

Did You Know?
The name 'kill' is historical. The command sends many different signals—not just signals that terminate a process.

Pro Tip
Most Linux administrators use plain 'kill PID' first and only switch to 'kill -9 PID' if the process does not respond.

Answer Key (Instructor Only)
Q1: sleep 300
Q2: ps -ef | grep sleep (or use htop)
Q3: kill PID && ps -ef | grep sleep
Q4: sleep 300 && kill -9 PID && ps -ef | grep sleep
Q5: pwd

Homework
Practice creating and stopping several sleep processes until you are comfortable identifying PIDs and choosing the correct signal.

Next Day Preview
Tomorrow you'll learn the jobs command to manage background jobs started from your current terminal.